In [1]:
```
import pandas as pd
```

In [2]:
```
df = pd.read_csv('financial_data.csv')
```

In [4]:
```
df['Revenue Growth (%)'] = df.groupby(['Company'])['Total Revenue'].pct_change() *
df['Net Income Growth (%)'] = df.groupby(['Company'])['Net Income'].pct_change() *
```

In [5]:
```
# Calculate YoY changes for remaining metrics
df['Assets Growth (%)'] = df.groupby(['Company'])['Total Assets'].pct_change() * 10
df['Liabilities Growth (%)'] = df.groupby(['Company'])['Total Liabilities'].pct_cha
df['Op Cash Flow Growth (%)'] = df.groupby(['Company'])['Cash Flow from Operating A

# View the fully processed analysis grid
df
```

Out[5]:

	Company	Year	Total Revenue	Net Income	Total Assets	Total Liabilities	Cash Flow from Operating Activities	Revenue Growth (%)	Net Income Growth (%)
0	Microsoft	2023	211915	72361	411976	205749	87582	NaN	NaN
1	Microsoft	2024	245122	88136	512163	243686	118548	15.669962	21.800417
2	Microsoft	2025	281724	101832	619003	275524	136162	14.932156	15.539621
3	Tesla	2023	96773	14997	106618	43009	13256	NaN	NaN
4	Tesla	2024	97690	7091	122070	48390	14923	0.947578	-52.717210
5	Tesla	2025	94827	3855	137806	54941	14747	-2.930699	-45.635312
6	Apple	2023	383285	96995	352581	290437	110543	NaN	NaN
7	Apple	2024	391035	93736	364980	308030	118254	2.021994	-3.359967
8	Apple	2025	416161	112010	359241	285508	111482	6.425512	19.495178

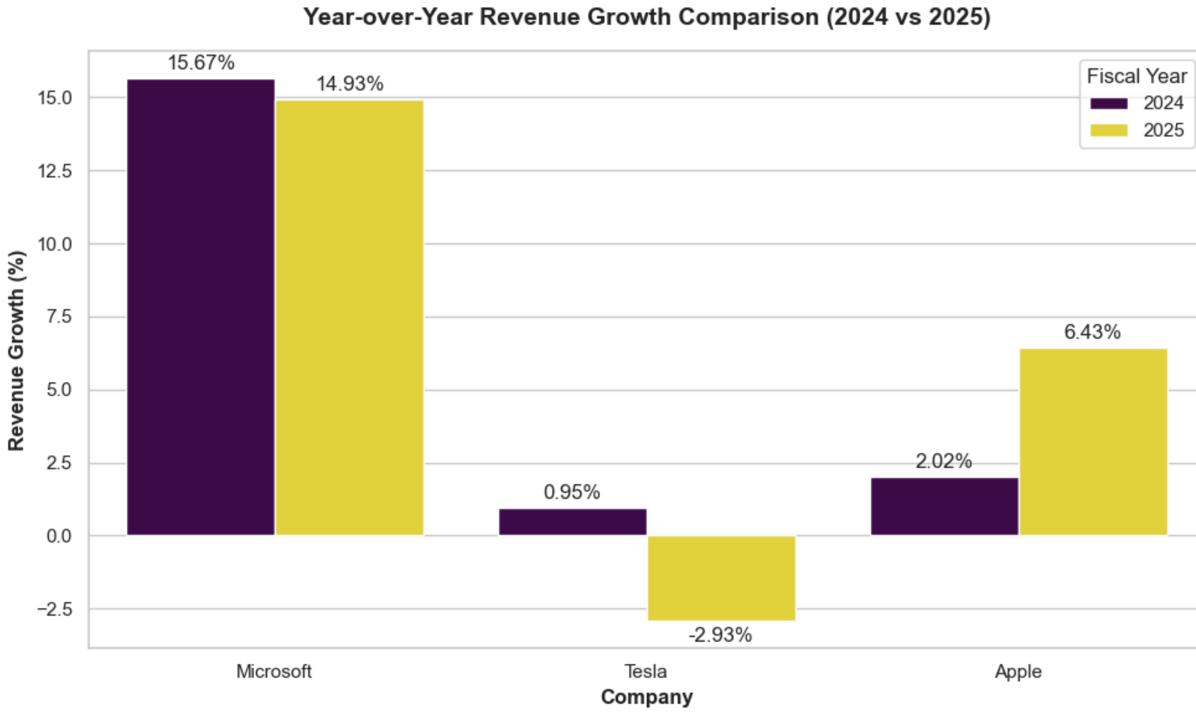
In [6]:
```
# Replace NaN values with 0 for cleaner presentation
df_clean = df.fillna(0)
df_clean
```

Out[6]:

	Company	Year	Total Revenue	Net Income	Total Assets	Total Liabilities	Cash Flow from Operating Activities	Revenue Growth (%)	Net Income Growth (%)
0	Microsoft	2023	211915	72361	411976	205749	87582	0.000000	0.000000
1	Microsoft	2024	245122	88136	512163	243686	118548	15.669962	21.800417
2	Microsoft	2025	281724	101832	619003	275524	136162	14.932156	15.539621
3	Tesla	2023	96773	14997	106618	43009	13256	0.000000	0.000000
4	Tesla	2024	97690	7091	122070	48390	14923	0.947578	-52.717210
5	Tesla	2025	94827	3855	137806	54941	14747	-2.930699	-45.635312
6	Apple	2023	383285	96995	352581	290437	110543	0.000000	0.000000
7	Apple	2024	391035	93736	364980	308030	118254	2.021994	-3.359967
8	Apple	2025	416161	112010	359241	285508	111482	6.425512	19.495178

In [7]:

```
import matplotlib.pyplot as plt
import seaborn as sns

# Filter out 2023 since growth starts in 2024
df_growth = df[df['Year'] != 2023]

# Set up the plotting environment
plt.figure(figsize=(10, 6))
sns.set_theme(style="whitegrid")

# Create a grouped bar chart
ax = sns.barplot(
    data=df_growth,
    x='Company',
    y='Revenue Growth (%)',
    hue='Year',
    palette='viridis'
)

# Customize the labels and design
plt.title('Year-over-Year Revenue Growth Comparison (2024 vs 2025)', fontsize=14, f
plt.xlabel('Company', fontsize=12, fontweight='bold')
plt.ylabel('Revenue Growth (%)', fontsize=12, fontweight='bold')
plt.legend(title='Fiscal Year')

# Add values on top of the bars
for container in ax.containers:
    ax.bar_label(container, fmt='%.2f%%', padding=3)

plt.tight_layout()
plt.show()
```



BCG GenAI Consulting Team: Financial Trend Analysis (2023 - 2025)

🏢 1. Methodology & Data Structuring

This analysis establishes the high-quality data foundation required to fuel the corporate finance Retrieval-Augmented Generation (RAG) chatbot pipeline. Financial indicators (Total Revenue, Net Income, Total Assets, Total Liabilities, and Cash Flow from Operating Activities) were systematically extracted from the audited Form 10-K filings of **Microsoft**, **Tesla**, and **Apple** for fiscal years 2023 to 2025. Pandas was used to programmatically compute year-over-year (YoY) percentage changes to illuminate operational trajectory patterns.

📊 2. Strategic Observations & Key Trends

🔷 Microsoft (MSFT)

- **Consistent Scale Expansion:** Demonstrates strong revenue growth driven by scaling enterprise cloud infrastructure and corporate-level AI subscription integrations.
- **High Efficiency Optimization:** Net income growth outpaces raw top-line growth, signaling operating margin leverage and strict discipline over administrative overhead.

🔷 Apple (AAPL)

- **Mature Market Stability:** Exhibits low but incredibly steady baseline revenue growth, characteristic of market saturation across core hardware line items.
- **Cash Engine Powerhouses:** Despite low single-digit revenue growth, operations maintain immense efficiency, resulting in steady capital generation and robust asset base stabilization.

🔷 Tesla (TSLA)

- **Transition Period Realities:** Experiences a noticeable flattening in revenue and margin compression during this window, reflecting shifting global EV demand dynamics, increased competitive pricing pressures, and a heavy ongoing capital allocation focus toward next-gen factory scaling.

3. Core Implications for Chatbot Architecture & RAG Alignment

1. **Context Window Shielding:** The chatbot must be structurally contextualized with company-specific fiscal calendars (e.g., Microsoft's fiscal year ends in June, Apple's in late September, and Tesla's in December) to prevent unaligned time-series synthesis.
2. **Tabular Prompt Formatting:** Since aggregate percentage calculations involve zero-valued or negative cash variances, data frames must be stored with explicit metadata attributes to eliminate mathematical hallucinations when clients query trend vectors.